

CRM Full-stack Agent Specification

This document outlines the requirements for building a complete Customer Relationship Management (CRM) system. It acts as a blueprint for the code-generation agent and details the database schema, backend API, business rules, user interface and design guidelines that must be followed. The goal is to replicate the functionality and look shown in the provided mock-ups while adding robust authentication and analytics.

Overview

Your task is to build a full-stack web application that allows real estate brokers (corretores) to manage their leads and interactions and allows managers (gestores) to monitor the productivity of their team. The system must include dashboards, follow-up scheduling, client management, interaction logging, KPI calculations and a modern **glassmorphism**-inspired user interface. The solution should be responsive and work on both desktop and mobile devices.

Platform

- **Database:** Use **PostgreSQL** with the schemas defined below. Use `uuid` as the primary key type. Store all timestamps in UTC.
- **Backend:** Expose a **REST API**. You may choose a framework such as **FastAPI** (Python) or **Next.js API routes** (TypeScript) to implement the endpoints. Include automatic OpenAPI/Swagger documentation.
- **Frontend:** Build a Single Page Application using **React** (with Next.js) or **Vue**. The frontend should consume the backend API and implement all dashboards and forms. Use **JWT** for authentication and store tokens in secure cookies or local storage.
- **Styling:** Implement the futuristic **glassmorphism** aesthetic—dark backgrounds, translucent panels with blurred backgrounds, rounded corners and neon accent colors. Use a CSS framework like **Tailwind CSS** to simplify layout and responsiveness.
- **Telephony & WhatsApp:** When displaying a phone number, wrap it in a `tel:` link so the device can initiate a call. Provide a button or link for WhatsApp that opens `https://wa.me/<E164_phone>` in a new tab.

Database Schema

Use the following tables and constraints in your PostgreSQL database. Create a `user_role` enum with values `'ADMIN'`, `'MANAGER'`, `'BROKER'`. Default timestamps use `now()` and all IDs are generated with `gen_random_uuid()`.

```
-- clients
CREATE TABLE public.clients (
  id uuid DEFAULT gen_random_uuid() PRIMARY KEY,
  name varchar NOT NULL,
  phone varchar NOT NULL,
  email varchar,
  source varchar NOT NULL,      -- e.g. "Google Ads", "Facebook",
```

```

"Indicação", "Website"
  status varchar NOT NULL,           -- lead status (see Status Workflow)
  owner_id uuid,                     -- FK to users.id (broker)
  observations text,
  created_at timestamptz DEFAULT now(),
  updated_at timestamptz DEFAULT now(),
  product varchar(255),
  property_value numeric(15,2),
  follow_up_state varchar(20) DEFAULT 'Sem Follow Up' CHECK (
    follow_up_state IN ('Ativo', 'Concluido', 'Pendente', 'Falhou', 'Sem Follow
Up')
  )
);

-- users
CREATE TABLE public.users (
  id uuid DEFAULT gen_random_uuid() PRIMARY KEY,
  name varchar NOT NULL,
  email varchar NOT NULL UNIQUE,
  password_hash varchar NOT NULL,
  role public.user_role DEFAULT 'BROKER' NOT NULL,
  created_at timestamptz DEFAULT now()
);

-- interactions
CREATE TABLE public.interactions (
  id uuid DEFAULT gen_random_uuid() PRIMARY KEY,
  client_id uuid NOT NULL REFERENCES public.clients(id) ON DELETE CASCADE,
  user_id uuid NOT NULL REFERENCES public.users(id) ON DELETE CASCADE,
  type varchar(255) NOT NULL,        -- e.g. "Ligação Iniciada", "Ligação
Registrada", "Mudança de Status"
  observation text,
  from_status varchar(255),
  to_status varchar(255),
  created_at timestamptz DEFAULT now(),
  updated_at timestamptz DEFAULT now()
);

```

Status Workflow

Define the following lead statuses for the `clients.status` column:

Status	Description
Primeiro Atendimento	New lead, first contact not yet made.
Fluxo de Cadência	Automated sequence of follow-ups.
Tratativa	Negotiation phase.
Aguardando Doc	Waiting for client documentation.

Status	Description
Doc Completa	All required documentation received.
Em Análise	Under review by the financial institution.
Aprovado	Approved for the purchase.
Reprovado	Not approved for purchase.
Venda Gerada	Deal closed and sale generated.
Arquivado	Archived lead (no longer active).

The follow-up state stored in `clients.follow_up_state` can be:

State	When it applies
Sem Follow Up	Client has never had a follow-up scheduled.
Ativo	A follow-up is scheduled for a future date/time.
Pendente	The follow-up due date/time has passed and it hasn't been concluded yet.
Concluido	The broker completed the follow-up on or before the due date.
Falhou	The broker or client cancelled the follow-up or it was unsuccessful.

Use triggers or background jobs to automatically update `follow_up_state` from `Ativo` to `Pendente` when the due date elapses.

Authentication and Authorization

- Implement **basic registration and login**. Passwords must be hashed using `bcrypt` before storage (store in `users.password_hash`).
- Issue **JWT tokens** upon successful login. Include the user ID and role in the token payload. Use middleware/guards to protect routes based on roles (`BROKER`, `MANAGER`, `ADMIN`).
- Only admins can create or manage users. Brokers can manage their own clients and view their personal dashboard. Managers can view aggregated metrics across all brokers.

API Endpoints

Design clean RESTful endpoints. Use proper HTTP methods and status codes. Below is a suggested API surface; adjust naming as needed:

Authentication

- `POST /auth/register` – Register a new user. Only accessible to admins. Body: `{ name, email, password, role }`.
- `POST /auth/login` – Authenticate a user and return a JWT token. Body: `{ email, password }`.

Users

- GET /users/me – Return the profile of the logged-in user.
- GET /users – List all users (manager/admin only).

Clients

- GET /clients – List clients. Accept query parameters:
- q – search text (matches name, phone or observations).
- status – filter by one of the lead statuses.
- source – filter by lead source.
- page, pageSize – pagination parameters.
- POST /clients – Create a new client. Body: { name, phone, email?, source, product?, property_value?, observations? }. Set owner_id to the current broker. Default status: 'Primeiro Atendimento'.
- GET /clients/{id} – Return details of a client, including current status, follow-up information and list of interactions.
- PUT /clients/{id} – Update client details (name, phone, email, source, product, property_value, observations). Only the owner or managers can update.
- DELETE /clients/{id} – Archive or delete a client. Use soft deletion or update status to 'Arquivado'.
- POST /clients/{id}/follow-up – Schedule a follow-up. Body: { due_date: timestamp, note? }. Set follow_up_state to 'Ativo' and record an interaction of type 'Agendar Follow Up'.
- POST /clients/{id}/status – Change lead status. Body: { to_status }. Record an interaction with type 'Mudança de Status' and fill from_status and to_status.
- POST /clients/{id}/complete-follow-up – Conclude a follow-up. Body: { state } where state is 'Concluido' or 'Falhou'. Update follow_up_state accordingly and record an interaction of type 'Conclusão de Follow Up'.

Interactions

- GET /clients/{id}/interactions – List all interactions for a client, ordered by created_at descending.
- POST /clients/{id}/interactions – Add an interaction. Body: { type, observation? }. Possible values for type include:
- 'Salvamento Anotação' – for saving a free-text note.
- 'Ligação Iniciada' – call started.
- 'Ligação Registrada' – call completed; if the call resulted in a contact effective (CE), prefix the observation with "CE: ".
- 'Mudança de Status' – automatically recorded when changing statuses.
- 'Agendar Follow Up' – automatically recorded when scheduling a follow-up.
- 'Conclusão de Follow Up' – recorded when concluding a follow-up.

Analytics

Broker KPIs

- GET /analytics/broker-kpis – Return static KPIs for the logged broker (independent of date range):

- `followUpAtrasado` – count of clients where `follow_up_state= 'Pendente'` and the due date is in the past.
- `followUpFuturo` – count of clients where `follow_up_state= 'Ativo'`.
- `primeiroAtendimento` – count of clients where `status= 'Primeiro Atendimento'`.
- `totalLeads` – count of clients with `status != 'Arquivado'`.

Broker Productivity

- `GET /analytics/productivity?start=<date>&end=<date>` – Return productivity metrics for the logged broker within the date range (inclusive):
- `calls` – number of interactions where `type IN ('Ligação Iniciada', 'Ligação Registrada')`.
- `ce` – number of `Ligação Registrada` interactions whose `observation` begins with "CE" (case-insensitive).
- `tratativas` – number of interactions with `type= 'Mudança de Status'` and `to_status= 'Tratativa'`.
- `docs` – number of interactions with `to_status= 'Doc Completa'`.
- `vendas` – number of interactions with `to_status= 'Venda Gerada'`.

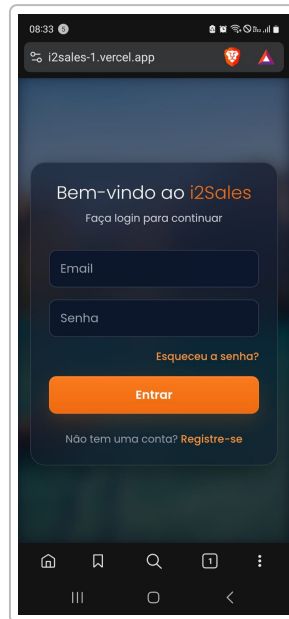
Manager Analytics

- `GET /analytics/manager?start=<date>&end=<date>&broker_id=<id|all>` – Return aggregated metrics for the specified broker or across all brokers:
- **VGV do Mês** (`vgvMes`) – sum of `property_value` for clients whose `status` changed to 'Venda Gerada' within the date range.
- **Oportunidades** – sum of `property_value` for clients currently in statuses 'Tratativa', 'Aguardando Doc', 'Doc Completa', 'Em Análise' during the date range.
- **calls, vendas** – sums of the corresponding broker productivity metrics.
- **temporalSeries** – daily counts of calls, CE, Tratativas, Docs, Vendas; include date and metric values for chart plotting.
- **funnel** – counts of unique clients that reached each stage: Calls, CE, Tratativa, Doc Completa, Vendas. The frontend will compute conversion percentages.
- **bySource** – aggregate table: for each `source`, provide counts of calls, CE, Tratativas, Docs, Vendas.
- **byBroker** – aggregate table: for each broker, provide the same counts.

Frontend Pages

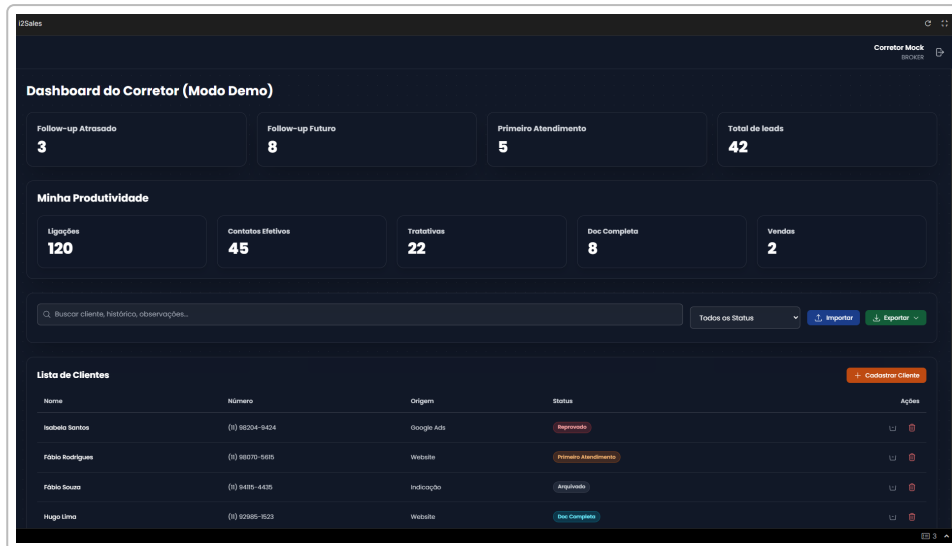
Login & Registration

- Provide a responsive login page similar to the mobile mock-up. Include fields for email and password and a button labelled "Entrar".
- Display a "Não tem uma conta? Registre-se" link if self-registration is allowed. Administrators can use a separate interface to create users.
- Example mobile view:



Broker Dashboard

- At the top, show four KPI cards: **Follow-up Atrasado**, **Follow-up Futuro**, **Primeiro Atendimento** and **Total de leads**. Each card displays a count and is clickable to filter the client list accordingly.
- Below, show a **"Minha Produtividade"** section with five dynamic KPIs: **Ligações**, **CE (Contatos Efetivos)**, **Tratativas**, **Doc Completa** and **Vendas**. Use a date range picker to filter these metrics; the range does not affect the static KPIs.
- Provide filters:
 - **Date range picker** – affects only productivity metrics.
 - **Search bar** – sends to `/clients?q=...`; matches on , and .
 - **Status dropdown** – sends to `/clients?status=...` to filter clients by their lead status.
- Display a **client list** with columns *Nome*, *Número*, *Origem*, *Status* and *Ações*. Provide a button labeled "Cadastrar Cliente" to open a modal or page for creating a client.
- Use the glassmorphism aesthetic: dark background, semi-transparent cards with blurred backgrounds, subtle gradient borders and bright accent colors. See the desktop view for guidance:

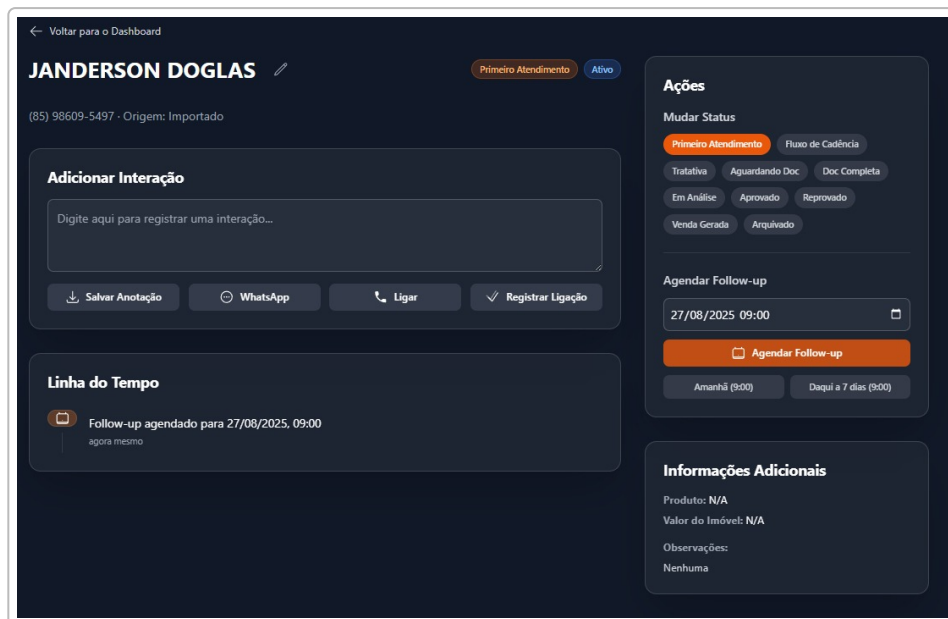


Client Detail & Interaction

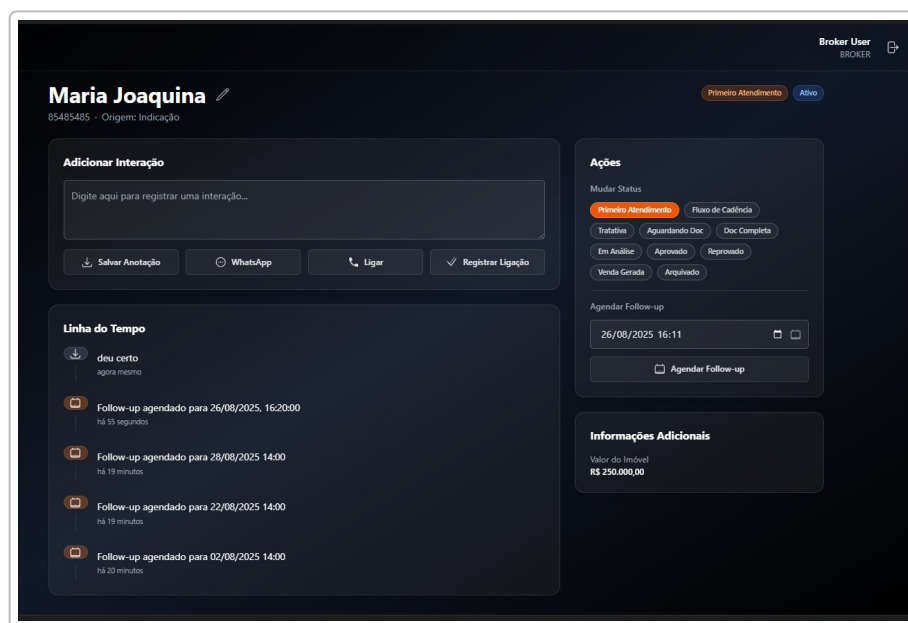
The client detail page is divided into three main sections:

1. **Header** – shows the client's name, phone (clickable `tel:` link), origin, a tag for current lead status and another tag for the follow-up state. Provide an edit button to modify the client name.
2. **Add Interaction** – provide a textarea for free-text notes. Below it, include buttons:
3. **Salvar Anotação** – posts an interaction with `type='Salvamento Anotação'`.
4. **WhatsApp** – opens `https://wa.me/<E164_phone>` in a new tab.
5. **Ligar** – opens `tel:<phone>`.
6. **Registrar Ligação** – opens a modal or directly posts an interaction with `type='Ligação Registrada'` and optional observation. If the call is a **Contado Efetivo**, prefix the observation with `"CE: "`.
7. **Ações** – allow the broker to change the lead's status via a list of clickable badges representing each status. When a status button is clicked, call `POST /clients/{id}/status`. Also provide a **Follow-up Scheduler**: a date/time picker and a button labeled "Agendar Follow-up". Predefined shortcuts ("Amanhã (09:00)", "Daqui a 7 dias (09:00)") fill the date picker.
8. **Linha do Tempo** – list the client's interactions in reverse chronological order. Use icons to represent notes, calls, status changes, follow-ups and follow-up completions. Show the time relative to now ("há 5 minutos", etc.) and the full observation text.

Example layout (desktop view):



And a mobile variation:

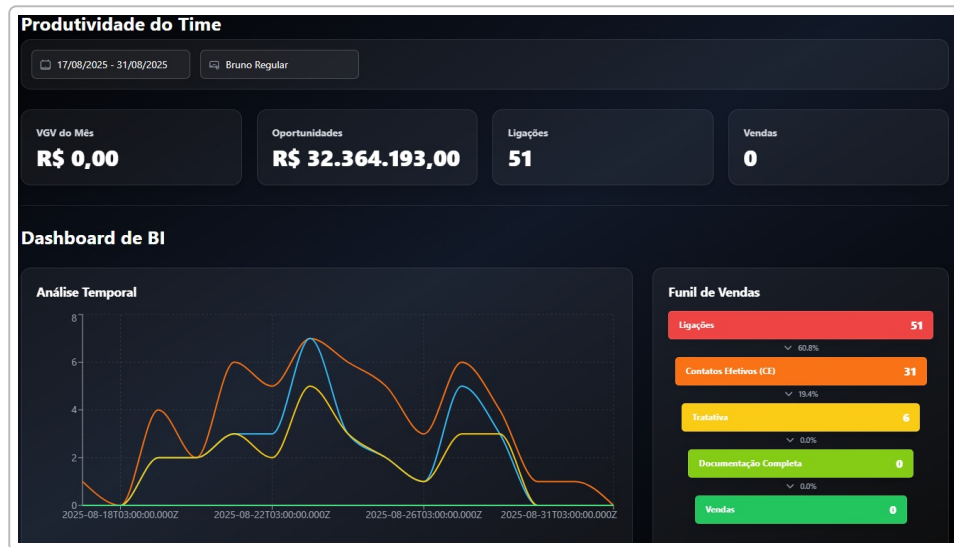


Manager Dashboard

- At the top, provide selectors for **Date Range** and **Broker** (with an option for "Todos"). When a broker is selected, the dashboard should update to show only that broker's data; when "Todos" is selected, aggregate data across all brokers.
- Display four KPI cards: **VGV do Mês**, **Oportunidades**, **Ligações** and **Vendas**. Values are computed based on the selected date range and broker.
- **Análise Temporal**: a multi-line chart that plots daily counts of *Ligações*, *CE*, *Tratativas*, *Doc Completa* and *Vendas* over the selected period. Use distinct colors for each line.
- **Funil de Vendas**: a horizontal bar or funnel diagram showing the number of leads that reached each stage (Calls, CE, Tratativa, Doc Completa, Vendas) and the conversion rate between stages.

Calculate the conversion as $(\text{count at stage N+1} / \text{count at stage N}) \times 100$. Display percentages as labels.

- **Detalhes por Origem / por Corretor:** two tables breaking down productivity metrics by lead source and by broker. Columns should include *Ligações*, *CE*, *Tratativas*, *Doc Completa* and *Vendas* for each group.
- Example look and feel:



Business Logic Summary

Broker KPIs (Static)

1. **Follow-up Atrasado** – count of clients assigned to the current broker where `follow_up_state='Pendente'` and the follow-up due date has already passed.
2. **Follow-up Futuro** – count of clients assigned to the broker where `follow_up_state='Ativo'` (a future follow-up is scheduled).
3. **Primeiro Atendimento** – count of clients assigned to the broker with `status='Primeiro Atendimento'`.
4. **Total de leads** – count of clients assigned to the broker where `status != 'Arquivado'`.

Broker Productivity (Date-dependent)

- **Ligações** – total interactions where `type IN ('Ligação Iniciada','Ligação Registrada')` within the selected date range.
- **CE (Contatos Efetivos)** – subset of `Ligação Registrada` interactions where the observation begins with `"CE"` (case-insensitive). This indicates a successful contact.
- **Tratativas** – number of status change interactions with `to_status='Tratativa'` during the date range.
- **Doc Completa** – number of status change interactions with `to_status='Doc Completa'` during the date range.
- **Vendas** – number of status change interactions with `to_status='Venda Gerada'` during the date range.

Manager KPIs (Aggregated)

Aggregate the same metrics across all brokers or for a single broker. Additional indicators include:

- **VGV do Mês** – sum of `property_value` for clients that moved to `status='Venda Gerada'` within the selected date range.
- **Oportunidades** – sum of `property_value` for clients currently in `status IN ('Tratativa', 'Aguardando Doc', 'Doc Completa', 'Em Análise')` during the date range.
- **Funil de Vendas** – counts of unique clients that progressed through each stage:
- **Ligações** – clients with at least one call.
- **CE** – clients with at least one effective contact.
- **Tratativa** – clients that reached the `Tratativa` status.
- **Documentação Completa** – clients that reached `Doc Completa`.
- **Vendas** – clients that reached `Venda Gerada`. The frontend should compute conversion rates between consecutive stages.
- **Detailed breakdowns** – for each `source` and each broker, provide counts for calls, CE, Tratativas, Doc Completa and Vendas.

Additional Notes

- Validate and sanitize all user input both on the client side and server side. Return meaningful error messages and HTTP status codes.
- When performing date calculations, consider the timezone **America/Fortaleza** (UTC-03:00). Store timestamps in UTC but convert to local time in the UI.
- Use pagination on lists and limit heavy analytics queries to avoid performance issues.
- Provide unit tests and integration tests to cover authentication, client CRUD, follow-up logic, status transitions and analytics.
- Document all endpoints with descriptions, parameters and responses in your API spec. If using **FastAPI**, the OpenAPI spec will be generated automatically.

Following this specification will allow the code-generation agent to produce a comprehensive CRM system that meets the functional requirements, matches the visual style of the mock-ups and correctly implements the complex business logic around follow-ups, status management and productivity analytics.
